

Software Project Management (SPM) (CSC415)

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Chapter 4

Risk Management

Risk

- Risk is any unexpected event that can affect your project — for better or for worse.
- Risk can affect anything: people, processes, technology, and resources.
- ***This is an important distinction: risks are not the same as issues.***
- Issues are things you know you'll have to deal with, and may even have an idea of when they'll occur, like a team member's scheduled vacation, or a big spike in product demand around the holidays.

Risk

- What is risk in project management?
 - Risks are events that *might happen, and you may not be able to tell when.*
- Risk is an uncertainty which is associated with a future and may or may not occur and a corresponding potential for loss.
- Two main characteristics of risk are:
 - ***Uncertainty***: the risk may or may not happen, that is there are no 100% risks.
 - ***Loss***: If risks become a reality then losses occur.

Types of Risk

- *Cost risk*, typically escalation of project costs due to poor cost estimating accuracy and scope creep.
- *Schedule risk*, the risk that activities will take longer than expected.
 - Slippages in schedule typically increase costs and, also, delay the receipt of project benefits, with a possible loss of competitive advantage.
- *Performance risk*, the risk that the project will fail to produce results consistent with project specifications.
- *Governance risk* relates to board and management performance with regard to ethics, community stewardship, and company reputation.
- *Strategic risks* result from errors in strategy, such as choosing a technology that can't be made to work.

Types of Risk

- *Operational risk* includes risks from poor implementation and process problems such as procurement, production, and distribution.
- *Market risks* include competition, foreign exchange, commodity markets, and interest rate risk, as well as liquidity and credit risks.
- *Legal risks* arise from legal and regulatory obligations, including contract risks and litigation brought against the organization.
- *Risks associated with external hazards*, including storms, floods, and earthquakes; vandalism, sabotage, and terrorism; labor strikes; and civil unrest.

Identify Risks

- Process of determining which risks may affect the project and documenting their characteristics.
- Key benefit of this process is the documentation of existing risks and the knowledge and ability it provides to the project team to anticipate events.
- It is an iterative process as risk can not be foreseen at once.
- SWOT Analysis (Strengths-Weakness-Opportunities-Threat Analysis) is a helpful technique for within greater organization to identify the risk.
- It is basically used to formulate the strategies of project for identifying risks and to find out weaknesses and threats.

Identify Risk: *Information Gathering Techniques*

- *Brainstorming:*

- The goal of brainstorming is to obtain a comprehensive list of project risks

- *Delphi technique:*

- The Delphi technique is a way to reach a consensus of experts.

- *Interviewing:*

- Interviewing experienced project participants, stakeholders, and subject matter experts helps to identify risks.

- *Root cause analysis:*

- Root-cause analysis is a specific technique used to identify a problem, discover the underlying causes that lead to it, and develop preventive action.

Identify Risk: *Diagramming Techniques*

- *Cause and effect diagrams:*
 - These are also known as fishbone diagrams and are useful for identifying causes of risks.
- *System or process flow charts:*
 - These show how various elements of a system interrelate and the mechanism of causation.
- *Influence diagrams:*
 - These are graphical representations of situations showing causal influences, time ordering of events, and other relationships among variables and outcomes.

Identify Risk: *Diagramming Techniques*

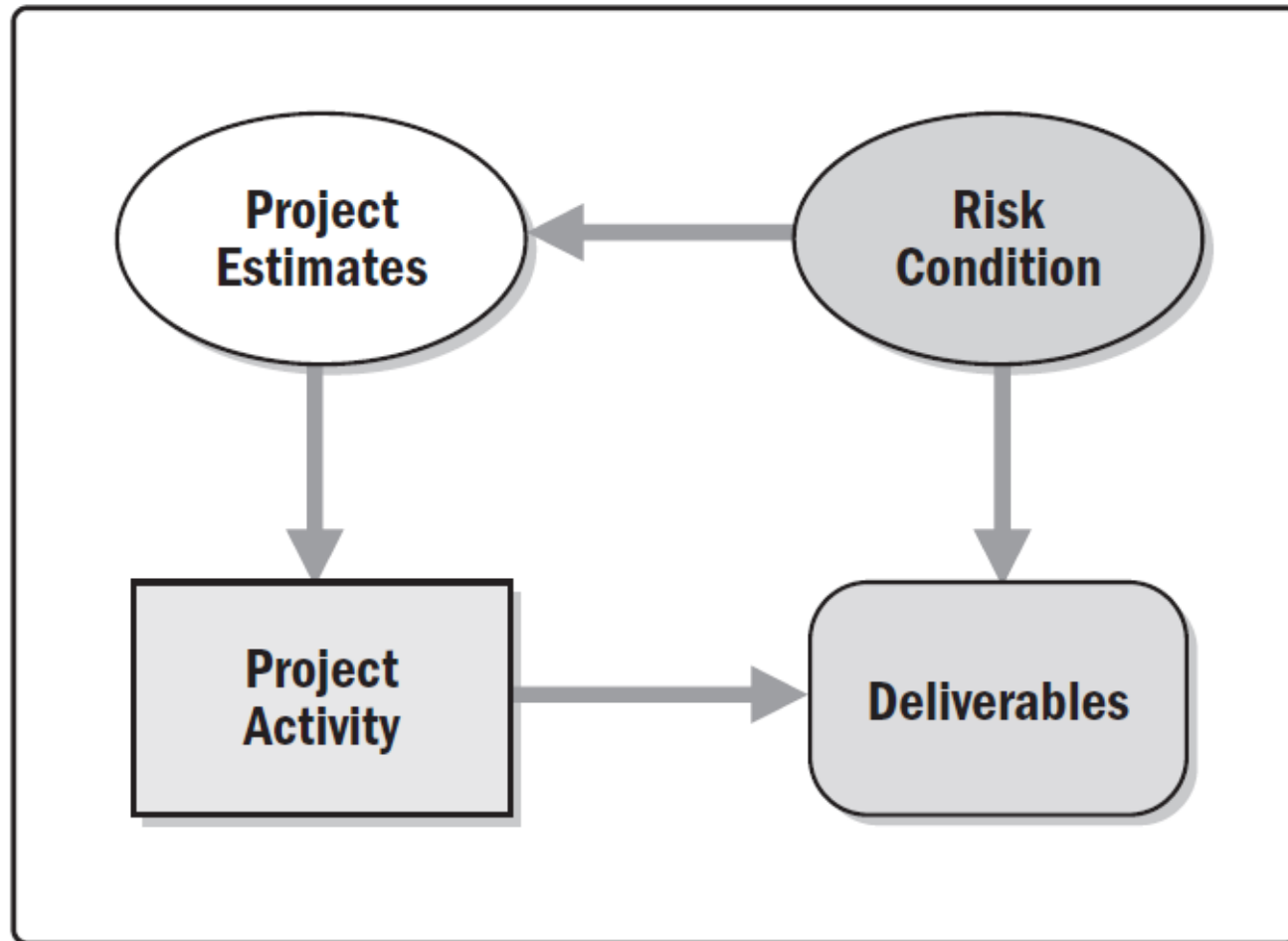


Figure 11-7. Influence Diagram

Risk Register

- Once the activities related to risk identification is completed, we can make the documentation about the risk called as risk register.
- The risk register is a document in which the results of risk analysis and risk response planning are recorded.
- It Includes:
 - *List of Identified Risks*
 - *List of Potential Responses*

Boehm's Top Ten Risk Items *(for Software Project)*

- *Personnel shortfalls*
- *Unrealistic schedules and budgets*
- *Developing the wrong software functions*
- *Developing the wrong user interface*
- *Gold plating*
- *Continuing stream of requirements changes*
- *Shortfalls in externally performed tasks*
- *Shortfalls in externally furnished components*
- *Real-time performance shortfalls*
- *Straining computer science capabilities*

Risk Analysis

- Risk analysis to in project management is a sequence of processes to identify the factors that may affect a project's success.
- This process includes risk identification (already discussed), analysis of risks and management of risks.
- It is a pro-active process which helps to control possible future events that may harm overall project.

Risk Analysis

- During risk analysis we perform the following steps:
 - ***Step_1:*** Identifying the problems causing risk in project.
 - ***Step_2:*** Identifying the probability occurrence of problems.
 - ***Step_3:*** Identifying the impact of problem.
 - ***Step_4:*** Assign value to step 2 and 3 range of 1-100 as (0-10) *very low*, (10-25) *low*, (25-50) *moderate*, (50-75) *high*, (75-100) *very high*

Risk Analysis

- **Step_5:** Calculate risk exposure factor as:

$$RE = (Potential\ Damage) \times (probability\ of\ occurrence)$$

i.e. **Step_3** * **Step_2**

Where, **potential value** can be a money value. For e.g.: flood caused damage of 15 crores.

Probability of occurrence ; ranges from 0.00 to 1.00 for e.g.: 0.1 (ten time in hundred)

- **Step_6:** Prepare table consisting of all of these values and order risk on the basis of risk exposure factor (RE).

Risk Analysis

PHASE X	PROBABILITY	IMPACT	RISK EXPOSURE
Risk X01	0.091	0.001	0.000
Risk X02	0.004	0.025	0.000
Risk X03	0.309	0.078	0.024
Risk X04	0.471	0.602	0.284
Risk X05	0.125	0.293	0.037

Risk Table with impact and probability of occurrence

Qualitative Risk Analysis

- It is the process of prioritizing risks for further analysis or action by assessing and combining their probability of occurrence and impact.
- It assesses the priority of identified risks using their relative probability or likelihood of occurrence, the corresponding impact on project objectives.
- It enables project managers to reduce the level of uncertainty and to focus on high-priority risks.
- Qualitative analysis of risk serves 3 functions:
 - Prioritize risks according to probability & impact
 - Identify the main areas of risk exposure
 - Improve understanding of project risks

Probability and Impact Matrix

Probability and Impact Matrix										
Probability	Threats					Opportunities				
0.90	0.05	0.09	0.18	0.36	0.72	0.72	0.36	0.18	0.09	0.05
0.70	0.04	0.07	0.14	0.28	0.56	0.56	0.28	0.14	0.07	0.04
0.50	0.03	0.05	0.10	0.20	0.40	0.40	0.20	0.10	0.05	0.03
0.30	0.02	0.03	0.06	0.12	0.24	0.24	0.12	0.06	0.03	0.02
0.10	0.01	0.01	0.02	0.04	0.08	0.08	0.04	0.02	0.01	0.01
	0.05/ Very Low	0.10/ Low	0.20/ Moderate	0.40/ High	0.80/ Very High	0.80/ Very High	0.40/ High	0.20/ Moderate	0.10/ Low	0.05/ Very Low

Impact (numerical scale) on an objective (e.g., cost, time, scope or quality)

Each risk is rated on its probability of occurring and impact on an objective if it does occur. The organization's thresholds for low, moderate or high risks are shown in the matrix and determine whether the risk is scored as high, moderate or low for that objective.

Figure 11-10. Probability and Impact Matrix

Quantitative Risk Analysis

- It is the process of numerically analyzing the effect of identified risks on overall project objectives
- It produces quantitative risk information to support decision making in order to reduce project uncertainty.
- It is performed on prioritized risks obtained after qualitative risk analysis.

Risk Avoidance and Mitigation

- Risk avoidance is a specific type of approach to managing risk, requiring a methodical process.
- The purpose of this technique is to altogether eliminate the occurrence of risks.
- So to avoid risks these method reduces the scope of projects by removing non-essential requirements.
- Leaders must identify and assess the risks their organization faces and determine how they will eliminate the chances of those risks causing damage to the organization.
- Because risk avoidance is a deliberate tactic, it is not the same as failing to identify a risk or ignoring it altogether.

Risk Avoidance and Mitigation

- Risk Mitigation refers to the strategy of planning and developing options to reduce threats to project objectives often faced by a business or organization.
- The goal of risk mitigation is not to eliminate threats.
- Rather, it focuses on planning for inevitable disasters and mitigating their impact on business continuity.
- Different types of potential risks include cyberattacks, natural disasters , legal liabilities, strategic management errors and accidents.

Risk Monitoring and Control Process

- Risk monitoring refers to an organization's framework for staying aware of its current risk exposure, including the implemented risk management system and any other activities that inform the organization's risk decisions.
- It is a key component of determining individual risk appetites – in other words, the decision of how much risk can be tolerated.
- More specifically, risk monitoring in an organization:
 - *Minimizes Risk*
 - *Mitigates the effects of Risk*
 - *Provides a clear picture of the Risk Landscape*
 - *Promotes Accountability*
 - *Creates Transparency*
 - *Utilizes Historical Events*
 - *Allows for Improvement*

Risk Monitoring and Control Process

- On the other hand controlling process means:
 - Process of implementing risk response plans
 - Tracking Identified Risks
 - Monitoring Residual Risks
 - Identifying New Risks
 - Evaluating Risk Process
- Improves efficiency of the risk approach throughout the project life cycle to continuously optimize risk responses.

Project Risk Management

- Includes the processes of conducting risk management planning, identification, analysis, response planning, and controlling risk on a project.
- Objectives are to increase the likelihood and impact of positive events, and decrease the likelihood and impact of negative events in the project.

Project Risk Management



Project Risk Management Processes

- *Plan Risk Management*—The process of defining how to conduct risk management activities for a project.
- *Identify Risks*—The process of determining which risks may affect the project and documenting their characteristics.
- *Perform Qualitative Risk Analysis*—The process of prioritizing risks for further analysis or action by assessing and combining their probability of occurrence and impact.

Project Risk Management Processes

- *Perform Quantitative Risk Analysis*—The process of numerically analyzing the effect of identified risks on overall project objectives.
- *Plan Risk Responses*—The process of developing options and actions to enhance opportunities and to reduce threats to project objectives.
- *Control Risks*—The process of implementing risk response plans, tracking identified risks, monitoring residual risks, identifying new risks, and evaluating risk process effectiveness throughout the project.

Project Risk Management Processes

- Project risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives such as scope, schedule, cost, and quality.
- Organizations perceive risk as the effect of uncertainty on projects and organizational objectives.
- Organizations and stakeholders are willing to accept varying degrees of risk depending on their risk attitude.

Project Risk Management Processes

- The risk attitudes may be influenced by a number of factors, which are broadly classified into three themes:
- *Risk appetite*, which is the degree of uncertainty an entity is willing to take on in anticipation of a reward.
- *Risk tolerance*, which is the degree, amount, or volume of risk that an organization or individual will withstand.
- *Risk threshold*, which refers to measures along the level of uncertainty or the level of impact at which a stakeholder may have a specific interest.
 - Below that risk threshold, the organization will accept the risk. Above that risk threshold, the organization will not tolerate the risk.

Plan Risk Management

- Process of defining how to conduct risk management activities for a project
- The risk management plan is vital to communicate with and obtain agreement and support from all stakeholders to ensure the risk management process is supported and performed effectively over the project life cycle.
- Careful and explicit planning enhances the probability of success for other risk management processes
- It is important to provide sufficient resources and time for risk management activities and to establish an agreed upon basis for evaluating risk

Risk Management Plan

- *Methodology:*

- Defines the approaches, tools, and data sources that will be used to perform risk management on the project.

- *Roles and responsibilities:*

- Defines the lead, support, and risk management team members for each type of activity in the risk management plan, and clarifies their responsibilities.

- *Budgeting:*

- Estimates funds needed, based on assigned resources, for inclusion in the cost baseline and establishes protocols for application of contingency and management reserves.

Risk Management Plan

- *Timing:*

- Defines when and how often the risk management processes will be performed throughout the project life cycle and establishes risk management activities for inclusion in the project schedule.

- *Risk categories:*

- Provide a means for grouping potential causes of risk.
- A risk breakdown structure (RBS) helps the project team to look at many sources from which project risk may arise in a risk identification exercise.

- *Definitions of risk probability and impact:*

- The quality and credibility of the risk analysis requires that different levels of risk probability and impact be defined that are specific to the project context.

Risk Management Plan

Table 11-1. Definition of Impact Scales for Four Project Objectives

Defined Conditions for Impact Scales of a Risk on Major Project Objectives (Examples are shown for negative impacts only)					
Project Objective	Relative or numerical scales are shown				
	Very low /0.05	Low /0.10	Moderate /0.20	High /0.40	Very high /0.80
Cost	Insignificant cost increase	< 10% cost increase	10 – 20% cost increase	20 – 40% cost increase	> 40% cost increase
Time	Insignificant time increase	< 5% time increase	5 – 10% time increase	10 – 20% time increase	> 20% time increase
Scope	Scope decrease barely noticeable	Minor areas of scope affected	Major areas of scope affected	Scope reduction unacceptable to sponsor	Project end item is effectively useless
Quality	Quality degradation barely noticeable	Only very demanding applications are affected	Quality reduction requires sponsor approval	Quality reduction unacceptable to sponsor	Project end item is effectively useless
This table presents examples of risk impact definitions for four different project objectives. They should be tailored in the Risk Management Planning process to the individual project and to the organization's risk thresholds. Impact definitions can be developed for opportunities in a similar way.					

Evaluation of Risk using Z-value

Example: The following table shows the jobs of a network along with their time estimates.

Activity	Estimated Duration (Days)		
	Optimistic (t_o)	Most Likely (t_m)	Pessimistic (t_p)
1-2	4	10	16
1-3	3	6	9
1-4	4	7	16
2-5	5	5	5
3-5	8	11	32
4-6	4	10	16
5-6	2	5	8

Evaluation of Risk using Z-value

Questions:

- *Draw network diagram.*
- *Identify the critical path and its duration.*
- *What is the probability that the project will be completed in 5 days earlier than the critical path duration?*
- *What project duration will provide 95% confidence level of completion?*

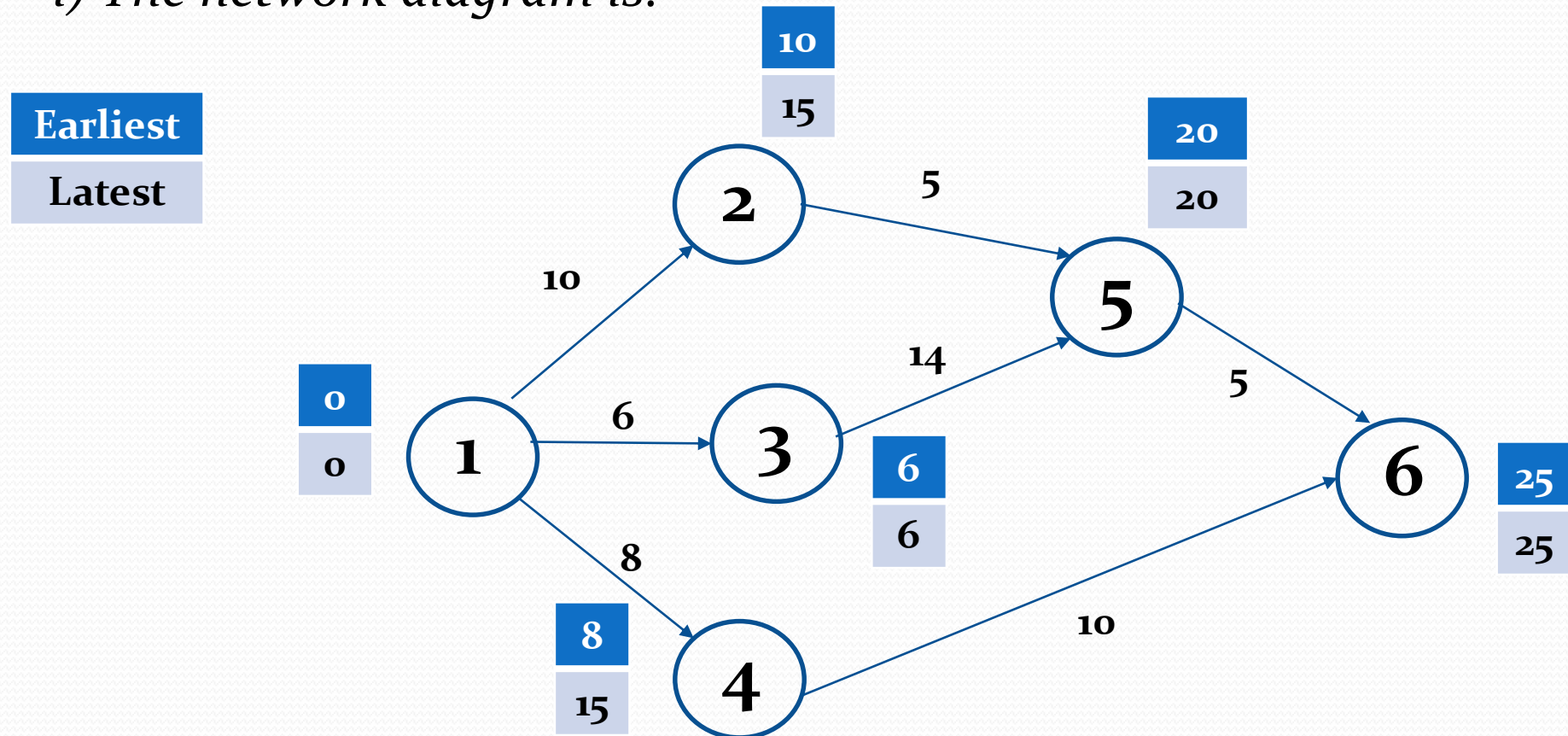
Solution,

The expected duration and variance is in the table below:

Activity	Estimated Duration (Days)			$t_e = \frac{t_o + 4t_m + t_p}{6}$	Variance (σ^2)= $((t_p - t_o)/6)^2$
	Optimistic (t_o)	Most Likely (t_m)	Pessimistic (t_p)		
1-2	4	10	16	10	4
1-3	3	6	9	6	1
1-4	4	7	16	8	4
2-5	5	5	5	5	0
3-5	8	11	32	14	16
4-6	4	10	16	10	4
5-6	2	5	8	5	1

Now we compute earliest and latest occurrence for each event by forward pass and backward pass same as we as we did for calculation of Early Finish and Late Finish on Critical Path Method.

i) The network diagram is:

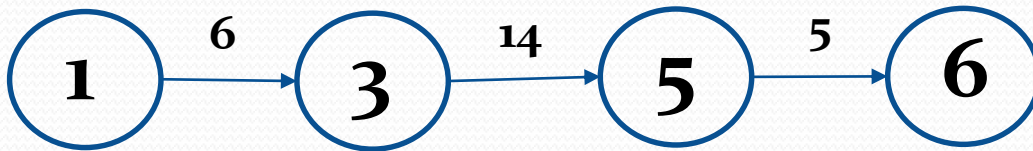


We, have several paths of this project

*To arrive at the **critical path**, we add the duration of each activity in the node at its given path.*

The critical path is the one with the longest duration.

*ii) **The longest path in this project is:***



*So the project length = **6+14+5**
= **25 days***

We need to add the variance of the critical path.

Since $1 \rightarrow 3 \rightarrow 5 \rightarrow 6$ is the critical path so $1 \rightarrow 3$, $3 \rightarrow 5$, $5 \rightarrow 6$ are the activities.

*So, add the values **1, 16, 1** since these are the respective variance of the activities in the critical path.*

Variance (σ^2) of project length = $\sum \sigma^2$ for critical path

$$= 1 + 16 + 1$$

$$= 18$$

So standard deviation (σ) = $\sqrt{18}$

$$= 4.24$$

iii) To Find the probability that the project will be completed in 5 days earlier than the critical path duration

We have critical path duration as 25 days.

*In question we need to find the probability that the project will be completed 5 days earlier **25 - 5 = 20 days**.*

The probability that the project will be completed in 20 days is given by $P(z \leq D)$, where $D = \frac{T_s - T_e}{\sigma}$

T_s is the schedule time i.e. 20 days.

T_e is the project length calculated earlier i.e. 25 days

$$D = \frac{20-25}{4.24} = -1.18$$

Now, $P(Z \leq -1.18) = 0.1190$
 $= 11.90\%$

Value from the probability
distribution table for -1.1 in .08
i.e. ninth column

Hence, the probability that the project will be completed in 20 days is 11.90%

iv) To Find project duration will provide 95% confidence level of completion

Given 95% confidence, so, from z-value table, the value for 0.95 (percent to decimal) is 1.65 (i.e. $z=1.65$).

So, using formula,

$$Z = \frac{T_s - T_e}{\sigma}$$

$$1.65 = \frac{T_s - 25}{4.24}$$

$$T_s = 31.97$$

Value from the probability distribution table where the value 0.95053 is in the row 1.6 and column 0.5 so the value is 1.65

Hence, Project duration for 95% level of confidence is 31.97 days



End of Chapter 4

Thank You !!!!